Data Dictionary & Privacy Statement

IT3091 Machine Learning — Industry Explorer Track

Group ID: `2026-AI-36`

Project: ChefCast AI — Restaurant Revenue & Product Demand Forecasting

Version: Final Submission — October 2026

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1. Raw Input Dataset Schema

File: `data/raw/pos_raw_dataset.csv`

Column Name	Data Type	Example Value	Description	Quality Notes
`INVDATE`	String → Date (YYYY-MM-DD)	`2025-03-15`	Transaction invoice date	Contains pre-2024 commissioning entries; filtered post-2024-03-27
`INVNO`	String (Alphanumeric)	`INV-045672`	Unique invoice (receipt) number	Used only for audit; 45,947 unique values
`ItemCode`	String	`F-0042`	Internal POS item catalog code	Not used in ML models; reference only
`ItemName`	String → Uppercase	`LUNCH`	Human-readable item name	Normalized to uppercase; 274 distinct values in raw data
`Qty`	Float	`3.0`	Number of units sold per line item	Contains negatives (voids); filtered to Qty > 0
`UnitPrice`	Float (LKR)	`350.0`	Price per unit in Sri Lankan Rupees	Contains extremes up to 2.41e11; filtered to ≤ 20,000 LKR

Derived column (not in raw file):

Derived Column	Formula	Description
`LineRevenue`	`UnitPrice × Qty`	Revenue contribution per transaction line item

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2. Processed Dataset Schemas

File: `data/processed/cleaned_pos_transactions.csv`

The audit-filtered transaction dataset after applying all 4 data quality filters.

Column	Type	Description
`Date`	Date (YYYY-MM-DD)	Transaction date (post-filter: 2024-03-27 to 2026-09-07)
`ItemName`	String	Uppercase-normalized menu item name
`Qty`	Float > 0	Net positive quantity sold
`UnitPrice`	Float (10–20,000 LKR)	Validated unit selling price
`LineRevenue`	Float	`UnitPrice × Qty`

Row count after cleaning: 125,912 (99.1% of original 127,093)

File: `data/processed/daily_revenue_series.csv`

Lens 1 target time series — aggregated daily revenue from the 11 core fresh items.

Column	Type	Description
`Date`	Date (YYYY-MM-DD)	Calendar date (895 continuous days)

`Revenue`	Float (LKR)	Total daily revenue from the 11 selected fresh handmade items

Row count: 895 (all calendar dates 2024-03-27 to 2026-09-07; zero-filled on non-trading days)

File: `data/processed/daily_product_demand_long.csv`

Lens 2 target panel series — long-format daily demand per dish.

Column	Type	Description
`Date`	Date (YYYY-MM-DD)	Calendar date
`ItemName`	String	One of 11 selected fresh handmade items
`Qty`	Float ≥ 0	Daily portion count sold for this item on this date

Row count: 9,845 (895 dates × 11 items; zero-filled on non-trading days per item)

File: `data/processed/daily_product_demand_wide.csv`

Wide-format version of the demand panel for visualization and EDA purposes.

Column	Type	Description
`Date`	Date	Calendar date
`LUNCH`	Float ≥ 0	Daily portions sold
`BREAKFAST`	Float ≥ 0	Daily portions sold
`CHICKEN_RICE_FULL`	Float ≥ 0	Daily portions sold
`CHICKEN_CURRY`	Float ≥ 0	Daily portions sold
`TEA`	Float ≥ 0	Daily portions sold
`ROLLS`	Float ≥ 0	Daily portions sold
`PATIS`	Float ≥ 0	Daily portions sold
`SAMOSA`	Float ≥ 0	Daily portions sold
`DATE_CAKE`	Float ≥ 0	Daily portions sold
`BUTTER_CAKE`	Float ≥ 0	Daily portions sold
`WATALAPPAN`	Float ≥ 0	Daily portions sold

File: `data/processed/product_selection_summary.csv`

Product scope justification summary from the 5-criterion selection analysis.

Column	Type	Description
`ItemName`	String	Product name
`TotalUnits`	Float	Total units sold across operational window
`TotalRevenue`	Float (LKR)	Total revenue contribution
`RevenueShare_pct`	Float	Percentage of total cleaned store revenue
`TradingDays`	Integer	Number of days with at least one sale
`MeanDailyQty`	Float	Average daily portion demand

3. Model Artifact Schema

Files: `models/lens1_*.joblib`, `models/lens2_*.joblib`

Serialized scikit-learn–compatible model objects (`.joblib` format, compatible with `joblib.load()`).

File	Model	Description

`lens1_LightGBM.joblib`	LightGBM Regressor	Lens 1 champion (retrained on Train+Val)
`lens1_Ridge_Regression.joblib`	Ridge Pipeline	Lens 1 Ridge model
`lens1_Random_Forest.joblib`	Random Forest	Lens 1 RF model
`lens1_XGBoost.joblib`	XGBoost	Lens 1 XGBoost model
`lens1_Stacking_Ensemble.joblib`	Custom TS Stacker	Lens 1 stacking ensemble
`lens1_Baseline_Seasonal_Naive.joblib`	Naive wrapper	Lens 1 naive baseline
`lens1_champion_model.joblib`	LightGBM (retrained)	Production deployment model for Lens 1
`lens2_Ridge_Regression.joblib`	Ridge Pipeline	Lens 2 champion (retrained on Train+Val)
`lens2_champion_model.joblib`	Ridge (retrained)	Production deployment model for Lens 2
`metrics.json`	JSON dict	Full benchmark metrics for both lenses

File: `models/metrics.json` — Schema

```
{ "lens1_revenue": { "<model_name>": { "validation": { "MAE": float, "RMSE": float, "WAPE_percent": float, "MAPE_percent": float, "R2_Score": float }, "test": { ... same fields ... } }, "champion_info": { "champion_model": string, "selection_method": "Lowest Validation WAPE", "validation_wape": float, "unbiased_test_wape": float, "empirical_residual_q90": float } }, "lens2_demand": { ... same structure ... } }
```

4. The 11 Selected Menu Items — Operational Dictionary

Item Name	Sri Lankan Culinary Context	Kitchen Station	Typical Prep Window	Shelf Life
`LUNCH`	Standard midday rice plate (rice, dhal, coconut sambol, seasonal vegetables, optional egg/fish curry)	Rice & Curry Station	09:00–11:00 AM	4–6 hours (must be served hot)
`BREAKFAST`	Morning set (string hoppers, pittu, or roti with coconut sambol, dhal, or egg curry)	Breakfast Station	05:30–08:00 AM	2–3 hours (best served fresh)
`CHICKEN RICE FULL`	Full-plate wok-fried rice with chicken, vegetables, egg, and soy seasoning	Wok/Rice Station	Prepared to order	1 hour (hot-held)
`CHICKEN CURRY`	Slow-simmered authentic Sri Lankan chicken curry with curry leaves, coconut milk, and aromatic spices	Curry Kitchen	06:00–10:00 AM (full batch cook)	6–8 hours (maintained hot)
`TEA`	Freshly brewed strong black Ceylon tea with optional milk and sugar	Beverage Station	Continuously brewed	30–45 minutes per pot
`ROLLS`	Handmade pastry rolls — thin crepe rolled around spiced minced filling, egg-washed, and deep-fried	Short-Eats Station	07:00–10:00 AM	2–3 hours (best consumed same day)
`PATIS`	Savory flaky pastry patties filled with potato, onion, and lentil mixture, baked or fried	Short-Eats Station	07:00–10:00 AM	3–4 hours
`SAMOSA`	Triangular pastry pockets filled with spiced potato-pea mixture, deep-fried until golden	Short-Eats Station	08:00–11:00 AM	2–3 hours

`DATE CAKE`	Traditional in-house baked moist date cake with caramel glaze — a Sri Lankan bakery specialty	Bakery Station	05:00–07:00 AM (daily bake)	24 hours (refrigerated)
`BUTTER CAKE`	Classic sponge butter cake baked daily with vanilla essence — standard bakery staple	Bakery Station	05:00–07:00 AM (daily bake)	24 hours (refrigerated)
`WATALAPPAN`	Traditional Malay-influenced steamed coconut jaggery custard pudding (kithul treacle, coconut milk, eggs, cardamom)	Dessert/Bakery Station	04:00–07:00 AM (steam batch)	6–8 hours (must be kept cool)

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5. Data Privacy & Responsible Data Handling Statement

5.1 Data Classification

Data Asset	Privacy Level	Handling
Raw POS transaction line items	Internal Business Data	Stored locally; not shared externally
Customer invoice numbers (`INVNO`)	Internal Reference	Used only for quality audit; not stored in model artifacts
Personal customer information	**None present**	The POS dataset contains no customer names, phone numbers, email addresses, payment card details, or biometric data
Product-level daily aggregates	Non-personal	Used as ML model training targets

5.2 Privacy Compliance Statement

The `pos_raw_dataset.csv` dataset does not contain any personally identifiable information (PII) as defined under:

- The **Personal Data Protection Act (PDPA)** of Sri Lanka (No. 9 of 2022)
- The **General Data Protection Regulation (GDPR)** principles (Article 4 — definition of personal data)

All ML model training, validation, and inference operates exclusively on product-level daily aggregate values. No customer-level information is processed, stored, or transmitted at any stage of the ChefCast AI pipeline.

5.3 Data Retention & Access Policy

- Raw POS dataset is retained only for the duration of the academic project.
- Model artifacts (`.joblib` files) contain no raw transaction data — only fitted model weights and coefficient vectors.
- Upon project submission, all datasets should be deleted from shared environments and stored only on the group's secured institutional storage.

5.4 AI-Use Transparency Acknowledgment

Generative AI tools (including large language models) were used for:

- Documentation draft generation and formatting assistance.
- Code syntax review and library function lookup.

All data analysis, model decisions, evaluation interpretations, and final report content were independently verified and validated by the student group members. No AI tool was used to fabricate results, generate synthetic data, or substitute for genuine analytical reasoning.